

Estimating bottom topography in shallow water flows

L. Pancotto^{1,2}, and P. Clark Di Leoni^{2,3}

¹Departamento de Física, FCEN, UBA, Buenos Aires, Argentina

²Universidad de San Andrés, Buenos Aires, Argentina

³CONICET, Argentina

Key Points:

- We develop two methods to estimate bottom topography in a shallow water flow using surface measurements.
- We test both methods using synthetic data
- Both methods are succesful in determining the bottom topography and surface velocity and are robust to noise

Corresponding author: Patricio Clark Di Leoni, pclarkdileoni@udesa.edu.ar

Abstract

We present two methods to estimate bottom topography in a shallow water flow using only surface deformation measurements. One is based on Physics-Informed Neural Networks (PINNs) and the other on the Adjoint State Method. We test both methods using synthetic data. Both are able to successfully reconstruct not only the bottom topography but also the surface velocity. Both also show robustness against noise and data sparsity up to reasonable levels.

Plain Language Summary

Determining what lies beneath the ocean surface is an open challenge. We present two techniques that can recover the shape of the bottom topography out of surface measurements. These techniques provide new ways to probe the depths of the oceans.

1 Introduction

The seafloor has always been a subject of speculation, captivating both imagination and intellect. What is that world lying beneath the bottom of our oceans like? More than 70% of the surface of our planet is covered by ocean waters, which amounts to approximately 362 million km² of the total surface area (Eakins & Sharman, 2007). Beyond the natural curiosity that arises when considering the shape of those vast landscapes, bathymetry plays a fundamental role in the countless activities and disciplines involving the Atmosphere and the Ocean. Understanding the shape of the seafloor is essential for modeling and comprehending ocean circulation (Gula et al., 2015), which in turn plays an important role in the predictive capacity of climate models for global phenomena (Santoso et al., 2011). Many current studies highlight the need to achieve a global mapping of the ocean floor, as well as the numerous difficulties such an endeavor entails (Wölfl et al., 2019; Mayer et al., 2018). Bathymetry directly affects the evaluation of geophysical risks of various kinds, such as landslides (Chiocci et al., 2011), or the formation and propagation of tsunamis (Degueldre et al., 2016). The topology of our ocean floors, after all, constitutes an essential boundary condition in the titanic problem of understanding the flows that govern the surface of our planet.

It is a fact that currently only a small portion of the area occupied by the oceans is charted (Smith et al., 2017). The vast task of documenting these topographies presents numerous obstacles. The earliest records of bathymetric measurements date back to Ancient Egypt, around 3000 years ago (Theberge, 1989). Bathymetry techniques available today include some that allow measurements from satellites—such as Satellite-Derived Bathymetry (SDB) (Shachak Pe’eri & Armstrong, 2014)—or from aircraft, like Light Detection and Ranging (LIDAR) (Irish & White, 1998). However, these are sensitive to water transparency and are generally used in shallow waters, where it is also riskier and more difficult to take measurements from vessels. Other strategies based on acoustic waves—such as Singlebeam Echo-Sounders (SBES) (Mayer, 2006) and Multibeam Echo-Sounders (MBES) (Glenn, 2015)—require seafaring, and involve higher costs and risks due to the need to navigate in the areas to be measured. Satellite Altimetry, alternatively, measures the height of the ocean surface from satellites and infers the seafloor topography from the gravitational effects it induces on the surface (Sandwell et al., 2006). While it allows for relatively fast surveying of the surface, it has a much lower resolution than bathymetry performed directly over the water.

In this work, we focus on studying bathymetry in flows under the Shallow Water (SW) approximation (Pedlosky, 1987) using data assimilation techniques. Specifically, this work compares the performance of Physics-Informed Neural Networks (PINNs) (Raissi et al., 2017) and the Adjoint State methodologies (ASM) (Dimet & Talagrand, 1986) in the task of inferring the topography of the bottom boundary of numerically sim-

ulated shallow water flows, assimilating data from only the height field. Flows governed by the SW equations have the advantage of eliminating vertical dependence by considering waves with wavelengths much greater than the fluid depth. Solving the proposed inverse problem then offers a tool for mapping the seafloor topography with surface measurements only. In nature, an example of flows with these characteristics are tsunamis. Modeling them this way is reasonable, as they typically have wavelengths of tens of kilometers and propagate in waters only a few kilometers deep. Far from the coast, the amplitude of these waves is in the order of one meter. The case study presented in (Degueldre et al., 2016) features a wavelength of $20km$ propagating in waters with an average depth of $4km$, with bottom fluctuations of a few hundred meters.

Physics-Informed Neural Networks (PINNs) are a recently introduced methodology that has shown significant advances (Cuomo et al., 2022; Cai, Mao, & Wang, 2021). They are based on supervised learning, as in traditional neural networks, but incorporate the constraints expressed by partial differential equations (PDEs) directly into the loss function (as residuals), thus enforcing the training process to satisfy the known physics of the system (Raissi et al., 2017). The cost function is minimized subject to both the measurements and the governing physical equations via gradient descent. This allows the inference of quantities that are not directly measured but appear in the equations, as is the case in this work for the bottom contour and velocity field. It also guides the training process to arrive at physically plausible solutions. This methodology offers several advantages — two important ones being the use of automatic differentiation to compute all required derivatives, and the fact that there is no need to define a discrete space-time grid for the problem domain (S. Wang et al., 2021). PINNs have been particularly successful at dealing with inverse problems such as the one presented here. For example, they have been used to reconstruct pressure fields from velocity measurements (Clark Di Leoni, Agarwal, et al., 2023), reconstruct velocity fields from temperature measurements (Clark Di Leoni, Agasthya, et al., 2023), reconstruct and enhance PIV measurements (Cai, Wang, et al., 2021; Hasanuzzaman et al., 2022).

The Adjoint State Method (ASM) is a variational approach to this problem. A cost function for the available field measurements is minimized subject to the physical equations of the problem. A Lagrangian is formulated that includes both the cost function from data to be extremized and the differential operators governing the physics. The adjoint problem must then be derived, specific to each cost function, and set of constraints in the form of differential operators; and finally solved (Dimet & Talagrand, 1986; Plessix, 2006). Solving the adjoint problem has a significantly lower computational cost than solving the problem directly via the Euler-Lagrange equations (Gaggioli & Bruno, 2022). The methodology was originally developed to perform data assimilation for numerical weather forecasting, but has been extended to other problems, such as reconstructing full three-dimensional turbulence (Li et al., 2020), locating scalar sources in turbulent flows (Q. Wang et al., 2019) and reconstructing various flows (M. Wang et al., 2019a; M. Wang & Zaki, 2021). In particular, (Du et al., 2023) compared both PINNs and a variational approach, finding the variational one to be more accurate and robust than PINNs. In this work, we present the derivation of the adjoint system of equations to apply it to the bathymetry problem in shallow water (SW), and the results of its application via pseudo spectral numerical integrators.

The objective of this work is to assess whether these methods are suitable for addressing the inverse problem of determining the bathymetry and velocity field by assimilating sparse surface measurements in SW flows. We compare their performance in cases with different sparsity for surface measurements, and different amplitudes of random noise added to the measurements (for a fixed sparsity). The manuscript is structured as follows. Section 2 derives the SW equations, discusses the pseudo spectral scheme utilized for numerical simulations, briefly discusses PINNs, present the theoretical framework we derived for the Adjoint State method in this problem, and introduces the parameters used

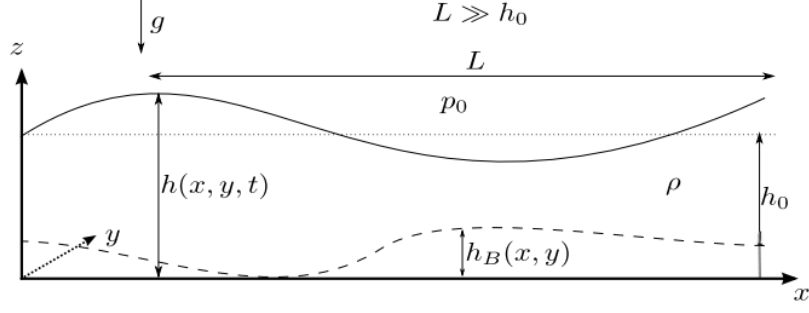


Figure 1: Diagram of the quantities involved in a SW problem. $h(x, t)$ is the fluid height field, $h_b(x)$ is the bottom topography, h_0 is the height of the fluid at rest, L is a characteristic horizontal distance, p_0 is the pressure just outside the fluid, ρ the fluid density, and g gravity.

for the numerical simulations and methods. In 4, we compare the results from both methods, in subsection 4.1 for cases with different sparsities in the data assimilated, and in subsection 4.2 for cases with different amplitudes of noise added to data. Section 5 presents the conclusions and future work prospects.

2 Equations and Methods

2.1 Shallow Water equations and problem formulation

Following the setup shown in Fig. 1 and starting from the Euler equations for an incompressible flow, assuming free surface boundary conditions on the surface and no-slip conditions at the bottom, and that the characteristic horizontal lengthscale L is much larger than the characteristic vertical lengthscale h_0 , one can derive the Shallow Water (SW) equations (Pedlosky, 1987) that describe the dynamics of the free surface

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} + g \nabla h = 0 \quad (1)$$

$$\frac{\partial h}{\partial t} + \nabla \cdot (\mathbf{u}(h - h_b)) = 0, \quad (2)$$

where $\mathbf{u} = (u, v)$ is the horizontal velocity field on the surface, h is the total fluid height, h_b is the bottom topography, and g is the acceleration of gravity. We also define the surface deformation $\eta = h - h_0$. For simplicity, we assume translational symmetry in the y -direction, thus removing all dependency on y and setting $v = 0$. All methods and results below follow this assumption, but they can be easily extended to the full two-dimensional case.

Under this model, the problem we try to solve is to estimate h_b given surface measurements $\{h_j\}_{j=1}^N$ at locations $\{x_j, t_j\}_{j=1}^N$. We can cast this task as an optimization problem where the goal is to find $\hat{u}$, $\hat{h}$ and $\hat{h}_b$ such that the cost function

$$\mathcal{L}_d = \frac{1}{N_d} \sum_{j \in \Omega_d} (h_j - \hat{h}_j)^2, \quad (3)$$

is minimized, subject to $\hat{u}$, $\hat{h}$ and $\hat{h}_b$ satisfying Eqs. (1) and (2) and where $\Omega_d = \{x_j, t_j; h_j\}_{j=1}^{N_d}$ is the aforementioned set of training data and $\hat{h}_j = \hat{h}(x_j, t_j)$. Below we show how this

problem can be solved with Physics-Informed Neural Networks and through the Adjoint State method.

2.2 Physics-Informed Neural Networks

Physics-Informed Neural Networks (PINNs) are neural networks whose training is regularized by the residuals of a system of differential equations of interest (Raissi et al., 2017; S. Wang et al., 2021), in this case the Shallow Water equations (1) and (2). We detail their implementation for the specific case of this work. Under this framework, neural networks, typically, fully connected multilayer perceptrons, are used to parametrize the relevant fields as a function of the coordinates. Here we use two separate networks. One parametrizes u and h and takes x and t as inputs, while the other parametrizes h_b and only takes x as an input. We refer to the output of the networks as $\hat{u}$, $\hat{h}$ and $\hat{h}_b$ and reserve the unhatted variables to the solutions of the SW equations, i.e., our ground-truth data.

Following the setup described in the previous section, the training process consists of optimizing the set of network parameters θ to minimize the cost function

$$\mathcal{L}_{\text{PINN}} = \mathcal{L}_d + \lambda_p(\mathcal{L}_u + \lambda_h \mathcal{L}_h), \quad (4)$$

where $\mathcal{L}_d$ is defined in Eq. (3), with $\{\hat{h}_j\}_{j=1}^N$ being the predictions of the PINN at the training points $\Omega_d = \{x_j, t_j\}_{j=1}^N$. The other two terms of the loss function are

$$\mathcal{L}_u = \frac{1}{N_p} \sum_{j \in \Omega_p} (\partial_t \hat{u}_j + \hat{u}_j \partial_x \hat{u}_j + g \partial_x \hat{h}_j)^2 \quad (5)$$

which are the residuals of Eq. (1), and

$$\mathcal{L}_h = \frac{1}{N_p} \sum_{j \in \Omega_p} (\partial_t \hat{h}_j + \partial_x (\hat{u}_j (\hat{h}_j - \hat{h}_{bj})))^2, \quad (6)$$

which are the residuals of Eq. 2. Note that these terms are evaluated at collocation points $\Omega_p = \{x_j, t_j\}_{j=1}^{N_p}$, which overlap, but are not equal to the points in Ω_d (more details below). Both networks share the same loss function, and their training is coupled via $\mathcal{L}_h$. All derivatives are calculated via automatic differentiation over the networks. The terms λ_p and λ_h are hyperparameters that balance the data and physics parts of the loss functions. In summary, the estimations $\hat{h}_b$ and $\hat{u}$ are incorporated only in (6) and (5), and optimized to satisfy the equations (1) and (2) only, while $\hat{h}$ is optimized to satisfy these equations as well as available data through $\mathcal{L}_d$. Further details on number of layer, hidden units, learning rates, balance hyperparameters λ_u and λ_h , and so forth are given below.

2.3 Adjoint State Method

In this section, we develop the formulation of the Adjoint State Method (ASM) for this problem, as an alternative to PINNs (Du et al., 2023; Plessix, 2006). The goal of this method is to find a way to calculate $\partial \mathcal{L}_d / \partial \hat{h}_b$ while enforcing the constraints of satisfying Eqs. (1) and (2) through the introduction of adjoint state variables, $\hat{u}^\dagger$ and $\hat{h}^\dagger$. We assume periodic boundary conditions over x for simplicity and ease of implementation and take 0 and T as the minimum and maximum times inside the assimilation window defined by Ω_d . The loss function reads

$$\mathcal{L}_{\text{adj}} = \mathcal{L}_d + \langle \hat{u}^\dagger, \partial_t \hat{u} + \hat{u} \partial_x \hat{u} + g \partial_x \hat{h} \rangle + \langle \hat{h}^\dagger, \partial_t \hat{h} + \partial_x (\hat{u} (\hat{h} - \hat{h}_b)) \rangle, \quad (7)$$

where $\langle \cdot, \cdot \rangle$ is an inner product defined over the whole spatio-temporal domain by

$$\langle \mathcal{A}, \mathcal{B} \rangle = \iint \mathcal{A} \mathcal{B} dx dt. \quad (8)$$

The optimality conditions are thus given by

$$\frac{\partial \mathcal{L}}{\partial \hat{h}_b} = 0, \quad (9)$$

$$\frac{\partial \mathcal{L}}{\partial \hat{u}} = 0, \quad \frac{\partial \mathcal{L}}{\partial \hat{h}} = 0, \quad (10)$$

$$\frac{\partial \mathcal{L}}{\partial \hat{u}^\dagger} = 0, \quad \frac{\partial \mathcal{L}}{\partial \hat{h}^\dagger} = 0. \quad (11)$$

The last two conditions are simply the SW equations, (1) and (2). The other conditions can be calculated by first using the chain rule on (7) to conjugate the inner products, obtaining

$$\begin{aligned} \mathcal{L}_{\text{adj}} = \mathcal{L}_d + \int_0^{2\pi} (\hat{u} \hat{u}^\dagger + \hat{h} \hat{h}^\dagger) \Big|_0^T dx \\ - \langle \hat{u}, \partial_t \hat{u}^\dagger + \partial_x (\hat{u} \hat{u}^\dagger) \rangle \\ - \langle \hat{h}, g \partial_x \hat{u}^\dagger + \partial_t \hat{h}^\dagger \rangle - \langle (\hat{h} - \hat{h}_b), \hat{u} \partial_x \hat{h}^\dagger \rangle, \end{aligned} \quad (12)$$

and then differentiating. Condition (9) yields

$$\frac{\partial \mathcal{L}}{\partial \hat{h}_b} = \frac{\partial \mathcal{L}_d}{\partial \hat{h}_b} - \langle -1, \hat{u} \partial_x \hat{h}^\dagger \rangle = 0, \quad (13)$$

and thus an expression for the gradient

$$\frac{\partial \mathcal{L}_d}{\partial \hat{h}_b} = - \int_0^T \hat{u} \partial_x \hat{h}^\dagger dt, \quad (14)$$

which is the term we are looking for. In order to calculate it we first need to calculate the adjoint fields, which can be obtained by solving conditions (17), which yield the equations for the evolution of the adjoint states

$$\partial_\tau \hat{h}^\dagger + \hat{u} \partial_x \hat{h}^\dagger + g \partial_x \hat{u}^\dagger = \sum_{j \in \Omega_d} 2(\hat{h}_j - h_j), \quad (15)$$

$$\partial_\tau \hat{u}^\dagger + \partial_x (\hat{u} \hat{u}^\dagger) + \hat{u} \partial_x \hat{u}^\dagger + (\hat{h} - \hat{h}_b) \partial_x \hat{h}^\dagger = 0. \quad (16)$$

Note that these equations are in reverse time τ , which goes from T to 0.

Since initial conditions $u(t=0)$, u_0 , and $h(t=0)$, h_0 , are unknown, optimality conditions must be imposed for them as well:

$$\frac{\partial \mathcal{L}_{\text{adj}}}{\partial \hat{u}_0} = 0, \quad \frac{\partial \mathcal{L}_{\text{adj}}}{\partial \hat{h}_0} = 0. \quad (17)$$

Imposing these conditions on boundary terms in equation (12) yields

$$\frac{\partial \mathcal{L}_d}{\partial \hat{u}_0} = \hat{u}^\dagger(x, t=0), \quad \frac{\partial \mathcal{L}_d}{\partial \hat{h}_0} = \hat{h}^\dagger(x, t=0). \quad (18)$$

These expressions give the direction to find the optimal $\hat{u}_0$ and $\hat{h}_0$ fields.

Since we assume $\mathcal{L}_d$ does not depend explicitly on $\hat{u}(t=T)$ and $\hat{h}(t=T)$, optimality conditions

$$\frac{\partial \mathcal{L}_{\text{adj}}}{\partial \hat{u}(t=T)} = 0, \quad \frac{\partial \mathcal{L}_{\text{adj}}}{\partial \hat{h}(t=T)} = 0 \quad (19)$$

hold if adjoint fields satisfy

$$\hat{u}^\dagger(\tau = 0) = 0, \quad \hat{h}^\dagger(\tau = 0) = 0. \quad (20)$$

These are the initial conditions for the inverse time evolution of the adjoint states.

The optimization problem to find $\hat{h}_b$ (and $\hat{h}(x, t = 0)$ and $\hat{u}(x, t = 0)$) can then be solved by implementing the following algorithm (M. Wang et al., 2019b):

1. Solve the SW system (1) and (2) using initial ansatz for $\hat{h}_b$, $\hat{h}(x, t = 0)$ and $u(x, t = 0)$ from $t = 0$ to the final simulation time $t = T$, storing the fields at each time step.
2. For the obtained fields $\hat{u}$ and $\hat{h}$, solve the adjoint system of equations defined by (16) and (15) backwards in time, from $\tau = 0$ to $\tau = T$, respecting the initial adjoint state conditions at $\tau = 0$. In this way, the adjoint state fields are obtained.
3. Calculate gradients (14) and (18), and use them in an optimization algorithm to update the values of $\hat{h}_b$, $\hat{u}_0$ and $\hat{h}_0$.
4. With the updated values, return to step 1 of the algorithm, and repeat until convergence.

The cost of calculating the gradient required for performing one step in the gradient descent using the adjoint state is equivalent to performing one *forward* integration of the SW equations, plus one *backward* integration of the adjoint state equations. Details on the numerical implementation and the optimization algorithm used are given below.

3 Numerical experiments

3.1 Numerical Simulations and Dataset Generation

One dimensional numerical simulations of Shallow Water flows were performed using a pseudo-spectral scheme to solve (1) and (2) in a 2π -periodical domain with 1024 grid point resolution. Numerical stability demands that simulation time step Δt satisfy the *CFL* condition, that is:

$$\Delta t < \frac{\Delta x}{c}, \quad (21)$$

being $c \sim \sqrt{g h_0}$ the characteristic velocity of wave propagation in the linearized case of (1) and (2) (Pedlosky, 1987). To eliminate spurious modes arising from aliasing, the 2/3 rule was applied: all modes satisfying $k > N/3$ were set to zero after performing each pseudo-spectral scheme to compute spatial partial derivatives of a nonlinear term. A second-order Runge-Kutta scheme was implemented for time integration.

We used a Gaussian wave packet as initial conditions

$$\begin{aligned} h(x, t = 0) &= h_0 + \eta_0 e^{-\left(\frac{x-x_0}{\sigma}\right)^2} \\ u(x, t = 0) &= \sqrt{\frac{g}{h_0}} \eta_0 e^{-\left(\frac{x-x_0}{\sigma}\right)^2}, \end{aligned} \quad (22)$$

and a bottom topography given by

$$h_b(x) = h_{b0}^{(1)} \exp\left(-\left(\frac{x-x_0^{(1)}}{\sigma^{(1)}}\right)^2\right) + h_{b0}^{(2)} \exp\left(-\left(\frac{x-x_0^{(2)}}{\sigma^{(2)}}\right)^2\right) + \sum_{j=0}^{60} A_j \sin(k_j x + \phi_j), \quad (23)$$

where wavenumbers k_j and phases ϕ_j were chosen at random from the uniform distributions $\mathcal{U}_k[40, 100]$ and $\mathcal{U}_\phi[0, \pi]$, and each amplitude A_j randomly chosen from values

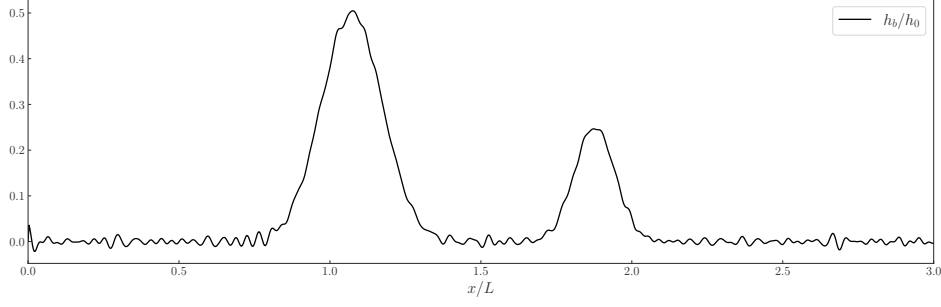


Figure 2: Bottom profile $h_b(x)$ used in the experiments.

g	h_0	η_0	σ	$\sigma^{(1)}$	$\sigma^{(2)}$	$x_0^{(1)}$	$x_0^{(2)}$	$h_{b0}^{(1)}$	$h_{b0}^{(2)}$
5	0.2	5×10^{-5}	$\frac{1}{2}$	0.3	0.2	$\frac{\pi}{1.4}$	$\frac{\pi}{0.8}$	0.1	0.05

Table 1: Parameters used to generate data.

n_x	1	16	64	128
δ_x	3/1024	3/64	3/16	3/8
l_x [m]	60	470	1900	3800

Table 2: Separation between measurements, in number of grid points n_x , normalized distances δ_x , and approximate physical distances l_x

between $1/4$ and $1/2$ the wavelength corresponding to each k_j . This profile is plotted in Figure 2. The values of the all the aforementioned parameters are given in Table 1.

When comparing quantities pertaining fields and distances from numerical simulations to the length of the simulated wave, that is $L = 2\pi/3$, we obtain relationships that are consistent with the SW hypothesis that L be larger than vertical quantities. For h , $h_0/L = 0.095$, and $\eta_0/L = 2.40 \times 10^{-5}$. For h_b , amplitudes satisfy $h_{b0}^{(1)}/L = 0.048$ and $h_{b0}^{(2)}/L = 0.024$. In nature, tsunamis are good candidate flows for being modeled by the Shallow Water equations. The width of the simulated wave can be mapped to the real physical scale of such a flow by defining that $\sigma \sim 10000m$ (Degueldre et al., 2016). Since the σ used yields a wave length approximately $L = 2\pi/3$, the mapping implies that the span of the 2π -periodic domain is $N \Delta x \sim 30000m$. This maps distance values in the simulation to physical units. Thus, the distance between grid points Δ_x is mapped to $29.3m$. The initial wave height is mapped to $\eta_0 \sim 1m$. The h_b main amplitudes are mapped to $h_{b0}^{(1)} \sim 2094m$ and $h_{b0}^{(2)} \sim 1047m$. The two main structures present in h_b will have their characteristic horizontal dimensions, $\sigma^{(1)}$ and $\sigma^{(2)}$, mapped to $6000m$ and $4000m$, respectively. The mean height from a flat bottom represents $h_0 \sim 4200$. The smaller structures present h_b are mapped to lengths $l \in [300m, 770m]$. Total simulation time is $T = 5$ with a time step $\Delta t = 1 \times 10^{-4}$. Time is normalized by $t_L = L/c$, which is the time for the propagating wave to travel one wave length. Thus, $T/t_L = 2.39$: just short of letting the wave propagate through the entire domain.

We perform experiments with different levels of sparsity in our simulated measurements. Table 2 show the different values used in terms of grid points, n_x , normalized distance $\delta_x = n_x \Delta_x / L$ and approximate physical distance (according to the values pre-

ϵ	1×10^{-7}	5×10^{-7}	1×10^{-6}	5×10^{-6}
ϵ/η_0	2×10^{-3}	1×10^{-2}	2×10^{-2}	1×10^{-1}

Table 3: Values of standard deviation ϵ for measurement noise, and relation to initial wave amplitude, ϵ/η_0 .

δ_x	3/1024	3/64	3/16	3/8
l	4	4	4	[4,6]
l_{inv}	4	4	4	4
w	100	100	100	[100,150,200]
w_{inv}	100	100	100	100

Table 4: PINN parameters for each case with distance δ_x . The brackets denote values used in different ensemble members for the sparsest case.

ϵ	1×10^{-7}	5×10^{-7}	1×10^{-6}	5×10^{-6}
l	[5,6]	[4,5,6]	[5,6]	[5,6]
l_{inv}	4	4	4	4
w	[100,150,200]	[100,150,200]	[100,150,200]	[100,150,200]
w_{inv}	100	100	100	100

Table 5: PINN parameters for each case with noise amplitude ϵ added to data. The brackets denote values used in different ensemble members for each case.

sented above), l_x . Additionally, for a fixed $\delta_x = 3/64$, noise is added to surface height data to test the robustness of each method. We use noise sampled from a Gaussian distribution $\mathcal{N}(0, \epsilon)$, with various standard deviations, ϵ that yield increasing amplitudes for the added noise. Table 3 shows the values of ϵ , along with their relation to the initial wave amplitude η_0 .

3.2 PINN parameters

The networks were trained using Adam with a mini-batch size of $m_b = 2000$, and a learning rate of $\mu = 1 \times 10^{-6}$. Each network was trained during 5990 epochs. The activation function used was *Siren* (Sitzmann et al., 2020). Different combinations of the number of layers l and hidden units w for the network that parametrizes u and h , and layers l_{inv} and hidden units w_{inv} for the network that parametrizes h_b , are indicated in table 4 for various experiments of different δ_x . For the sparsest case, that is for $\delta_x = 3/8$, an ensemble of PINNs with different combinations of layers and hidden units were used. Ensembles of PINNs were also used for the cases with added noise to measurements. The parameters used to the networks for these cases are indicated in table 5. In all other cases, that is $\delta_x = \{3/1024, 3/64, 3/16\}$, only one PINN is used, per case. The results shown represent the mean of the ensemble.

Following (S. Wang et al., 2021), the balance hyperparameters λ_u and λ_h were updated at each i -th epoch by weighing the gradients of the loss terms in (4) against each

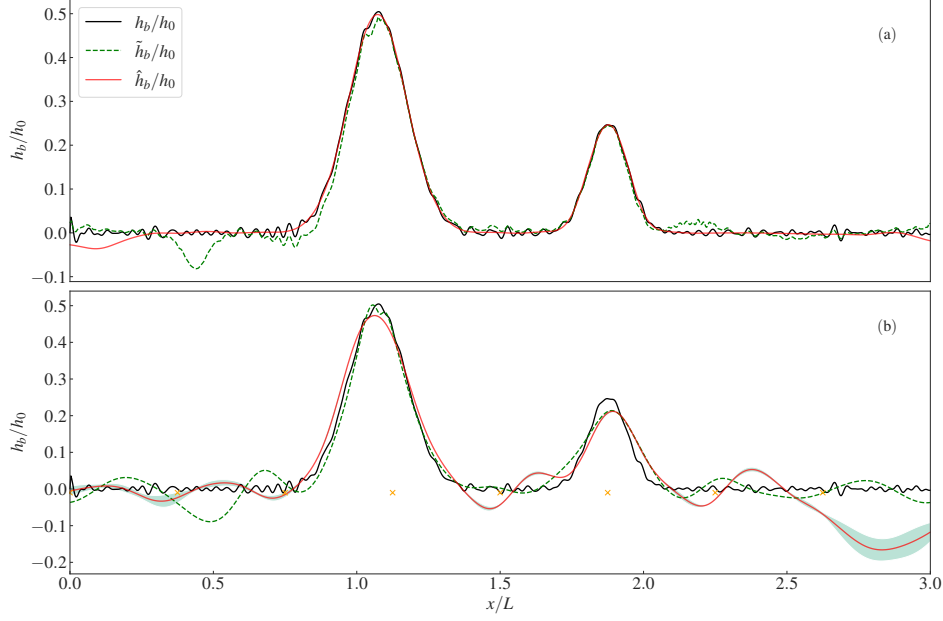


Figure 3: (a) Results for h_b field using a separation $\delta_x = 3/1024$ between measurements. $\hat{h}_b$ for PINN method (red solid line), $\tilde{h}_b$ ASM (green dotted line), and the true h_b (black solid line), plotted in the 2π -periodic domain normalized by L . (b) Same for $\delta_x = 3/8$. The blue shade represents the standard deviation between different PINN realizations, for the same measurements. The yellow crosses mark the measurement locations.

other:

$$\lambda_p^{i+1} = (1 - \alpha)\lambda_p^i + \alpha \frac{\langle \nabla_\theta L_d^i \rangle}{\langle \nabla_\theta L_u^i + \lambda_h^{i+1} \nabla_\theta L_h^i \rangle}, \quad (24)$$

where α was a tuneable hyperparameter, and where

$$\lambda_h^{i+1} = (1 - \alpha)\lambda_h^i + \alpha \frac{\langle \nabla_\theta L_u^i \rangle}{\langle \nabla_\theta L_h^i \rangle}, \quad (25)$$

so that the total gradient of (4) at epoch $i + 1$ yielded

$$\nabla_\theta \mathcal{L}^{i+1} = \nabla_\theta L_d^{i+1} + \lambda_p^{i+1} (\nabla_\theta L_u^{i+1} + \lambda_h^{i+1} \nabla_\theta L_h^{i+1}). \quad (26)$$

For physics loss terms relative weight tuning in (24) and (25), we used $\alpha = 0.1$.

3.3 Adjoint State Method

The adjoint state equations were solved using the same pseudo-spectral method as for the SW equations described above. The only caveat being that the forcing term was projected using Fourier interpolation so as to avoid discontinuities. The L-BFGS algorithm was used for the optimization steps. The gradient descent step was 10^{-1} for h_b field optimization, and 10^{-2} for initial fields $u(t = 0)$ and $h(t = 0)$.

The ansatz for the initial height $\hat{h}(t = 0)$ is built by interpolating the sparse measurements by fitting them to a Gaussian curve. This is done by optimizing the parameters A, B, x_0, σ in

$$B + A e^{-(\frac{x-x_0}{\sigma})^2}$$

to the measurements, using a least squares approach. We generate an ansatz for the initial velocity field by taking $\hat{u}_0 = \sqrt{g/h_0 \hat{\eta}_0}$.

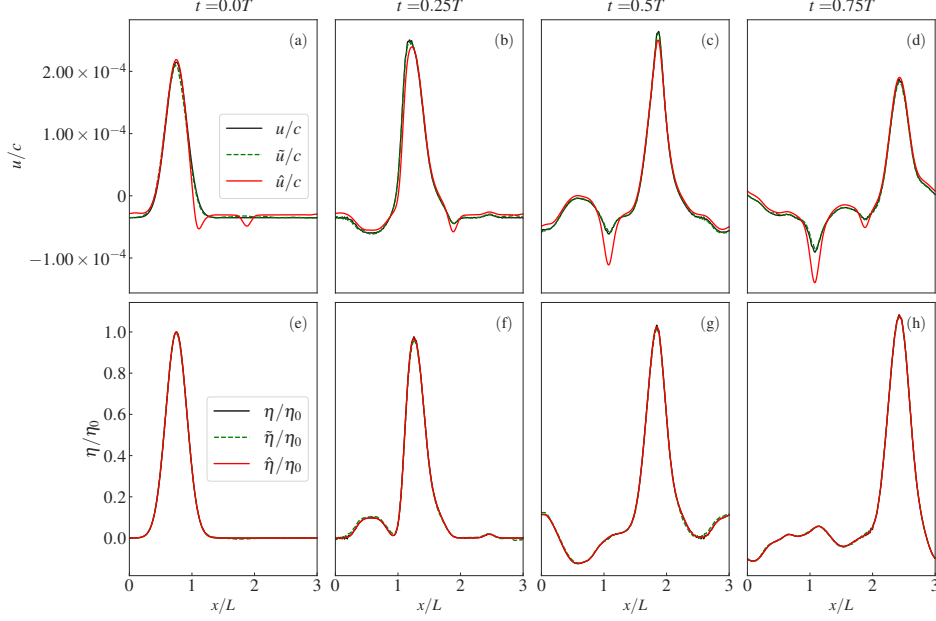


Figure 4: Dense measurement case, $\delta = 3/1024$. (a-d): true and reconstructed velocity fields u/c at different times. (e-h): true and reconstructed surface fluctuations η/η_0 . Same legends as figure 3.

3.4 Error Metrics

The global error for PINN predictions is defined as

$$\mathcal{E}_{h_b} = \frac{\int_0^{2\pi} \sqrt{(h_b - \hat{h}_b)^2} dx}{\int_0^{2\pi} \sqrt{(h_b)^2} dx}, \quad (27)$$

and

$$\mathcal{E}_u = \frac{\int_0^{2\pi} \int_0^\tau \sqrt{(u - \hat{u})^2} dt dx}{\int_0^{2\pi} \int_0^\tau \sqrt{(u)^2} dt dx}. \quad (28)$$

The global error for ASM predictions is defined as

$$\mathcal{E}_{h_b} = \frac{\int_0^{2\pi} \sqrt{(h_b - \tilde{h}_b)^2} dx}{\int_0^{2\pi} \sqrt{(h_b)^2} dx}, \quad (29)$$

and

$$\mathcal{E}_u = \frac{\int_0^{2\pi} \int_0^\tau \sqrt{(u - \tilde{u})^2} dt dx}{\int_0^{2\pi} \int_0^\tau \sqrt{(u)^2} dt dx}. \quad (30)$$

These will serve to make global comparisons between both methods in this work.

4 Results

4.1 Sparse measurements

We begin by showing the results for PINNs and ASM for the case without any added noise, $\epsilon = 0$, and for different distances between measurements δ_x . Figure 3 shows predictions from PINNs $\hat{h}_b$ (red solid line) and from the ASM $\tilde{h}_b$ (green dotted line), plotted against the true h_b field (black solid line) for (a) $\delta_x = 3/1024$ and (b) $\delta_x = 3/8$.

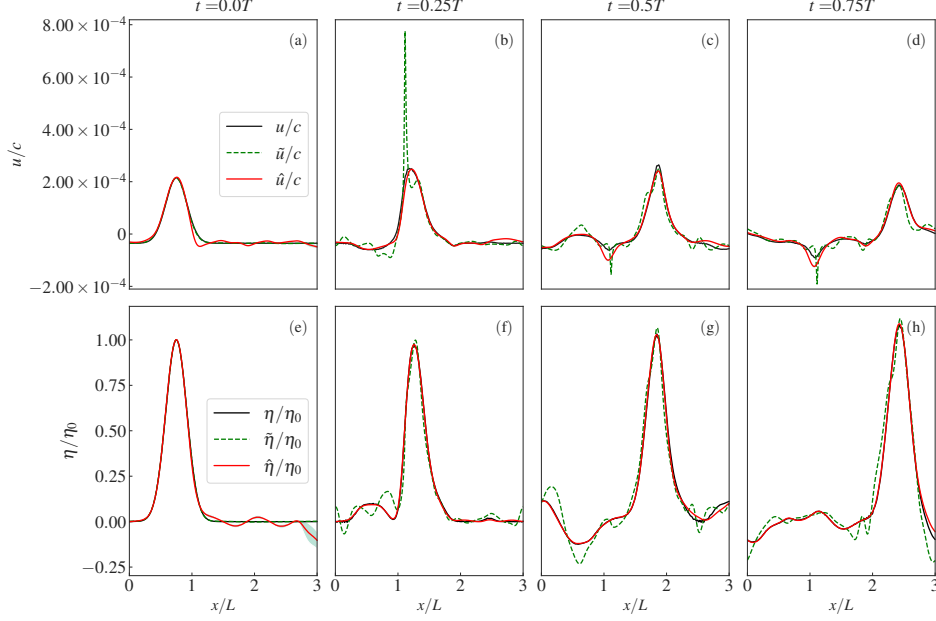


Figure 5: Sparse measurement case, $\delta = 3/8$. (a-d): true and reconstructed velocity fields u/c at different times. (e-h): true and reconstructed surface fluctuations η/η_0 . Same legends as figure 3.

Both methods can reconstruct the overall shape of the bottom topography with good accuracy, even in the sparser case. The ASM calculations took between 2 and 4 hours, the longest being the less sparse cases, where smaller scales could be resolved. Training the PINNs took from 10 to 24 hours to converge. The PINN ensemble shown in subfigure (b) is made out of 5 $\hat{h}_b$ predictions. Different network architectures and random initializations of the network parameters lead to different results after optimization. The mean of all PINN realizations for a single case is considered.

Figure 4 shows predictions for fields u and h , for the case with $\delta_x = 3/1024$. It is noticeable that, while both methods are mostly successful in reconstructing the the velocity field, the ASM is considerably better. As for the surface height, both methods are trivially successful in its reconstruction, since both have access to surface height data, in this case, with full spatial resolution. For the sparser case of $\delta_x = 3/8$, shown in figure 5, the ASM presents much larger fluctuations. Indeed, for some frames, such as figure 5(b) where the wave is traveling over the largest gradient for h_b , there appear to be discontinuities in $\tilde{u}$ while $\hat{u}$ remains quite smooth. It is noticeable as well that $\hat{h}$ outperforms $\tilde{h}$ for $\delta_x = 3/8$, as seen in frames (e) to (h) in figure 5.

Finally, figure 6 shows (a) global errors $\mathcal{E}[\hat{h}_b]$ and $\mathcal{E}[\tilde{h}_b]$ defined in equation (29), and (b) $\mathcal{E}[\hat{u}]$ and $\mathcal{E}[\tilde{u}]$ as defined in equation (30), for each sparsity in data points considered. In both methods the error in reconstructing the bottom topography increases as the separation between measurements increases, as expected, with PINNs outperforming ASM when measurements are dense and ASM outperforming PINNs when they are sparse. These results are reversed when looking at the errors of the reconstruction of the velocity fields in panel (b) where ASM achieves better results than PINN when measurements are dense. Interestingly, the error in the PINN reconstruction is stable with respect to measurement distance in this case.

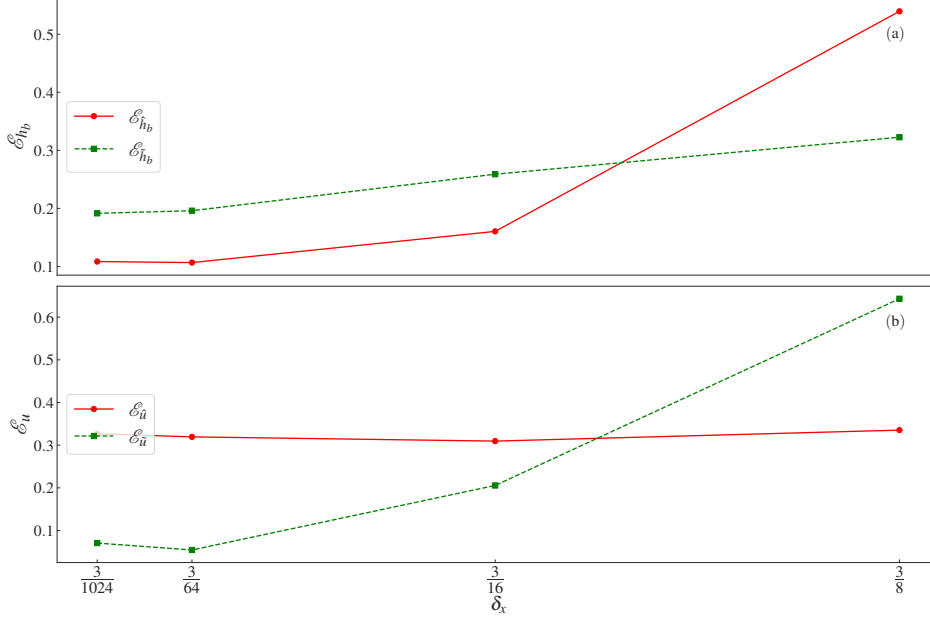


Figure 6: Errors as function of the measurement distance δ_x . (a) Global error $\mathcal{E}_{\hat{h}_b}$ (red solid line) and $\mathcal{E}_{\tilde{h}_b}$ (green dotted line), and (b) Global error $\mathcal{E}_{\hat{u}}$ (red solid line) and $\mathcal{E}_{\tilde{u}}$ (green dotted line).

4.2 Noisy measurements

This subsection discusses the results of PINNs and ASM when assimilating data with different amplitudes of added noise. In all cases presented here the distance between measurements is $\delta_x = 3/64$. Figure 7 shows the bottom topography reconstructed by the PINN, $\hat{h}_b$, and by the ASM, $\tilde{h}_b$ for the cases with (a) $\epsilon/\eta_0 = 0.002$ and (b) $\epsilon/\eta_0 = 0.1$. Both methods are able to reconstruct the main structures correctly, although PINNs achieves a more accurate result. It is interesting to note too how in the noisier case, the different PINN outcomes have much greater variance than in Figure 3.

The global error committed by predictions from both methods for each amplitude of noise added to data is plotted in figure 8, for (a) h_b and (b) u . Figure 8(a) shows a consistent scaling for both methods to noise in data, with a slightly more pronounced growth for $\mathcal{E}[\hat{h}_b]$. In figure 8(b) values remain relatively stable for both methods, except for an abrupt increase in $\mathcal{E}[\hat{h}_b]$ at $\epsilon/\eta_0 = 1 \times 10^{-1}$. Note that values of global error in figure 8 start close to what they are in the noiseless case, seen in figure 6.

5 Conclusions

In this work we presented two methods to estimate bottom topography using surface height data under the Shallow Water approximation. One approach is based on physics-informed neural networks (PINNs), while the other on the adjoint state method (ASM). Both are able to reconstruct the topography as well as the surface velocity field with good accuracy. In particular, we were able to reconstruct structures that about half the size of the length of the incoming wave, which should represent distances between 4 and 6km in the ocean. This is on par with current available methods. Results were robust as long as the distance between measurements was less than half the length of the incoming wave and the noise was less than 10%. PINNs produced overall smoother results and was no-

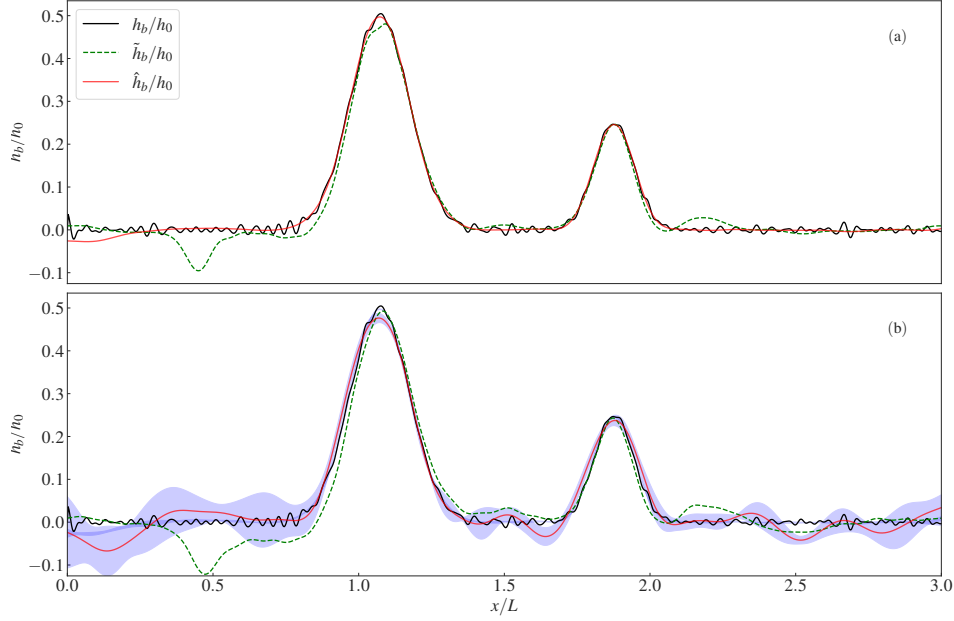


Figure 7: (a) Results for h_b field for noise amplitude $\epsilon/\eta_0 = 2 \times 10^{-3}$ and separation $\delta_x = 3/64$ between data points. $\hat{h}_b$ for PINN method (red solid line), $\tilde{h}_b$ ASM (green dotted line), and the true h_b (black solid line), plotted in the 2π -periodic domain normalized by L . (b) Results for h_b field for noise amplitude $\epsilon/\eta_0 = 1 \times 10^{-1}$ and separation $\delta_x = 3/64$ between data points. $\hat{h}_b$ for an ensemble of PINNs (red solid line), $\tilde{h}_b$ (green dotted line), and h_b (black solid line), plotted in the 2π -periodic domain normalized by L . The blue shade represents the standard deviation between different PINN realizations, for the same data points. The yellow crosses mark the measurement locations.

tably easier to implement. The results presented here are in one dimension but can be easily extended to two. In future work we will apply both methodologies against experimental measurements and observations.

Open Research Section

Codes can be found at <https://github.com/LucasPancotto/inverseSW>. The repository includes the codes used to generate the ground truth data, both the PINN and the ASM implementations of the reconstruction methods and all the scripts needed to plot the figures in the present manuscript.

Acknowledgments

The authors would like to thank Pablo Cobelli for useful discussions. The authors do not report any conflict of interests.

References

- Cai, S., Mao, Z., & Wang, Z. (2021). Physics-informed neural networks (pinns) for fluid mechanics: a review. *Acta Mechanica Sinica*, 37, 1727–1738. doi: <https://doi.org/10.1007/s10409-021-01148-1>
- Cai, S., Wang, Z., Fuest, F., Jeon, Y. J., Gray, C., & Karniadakis, G. E. (2021).

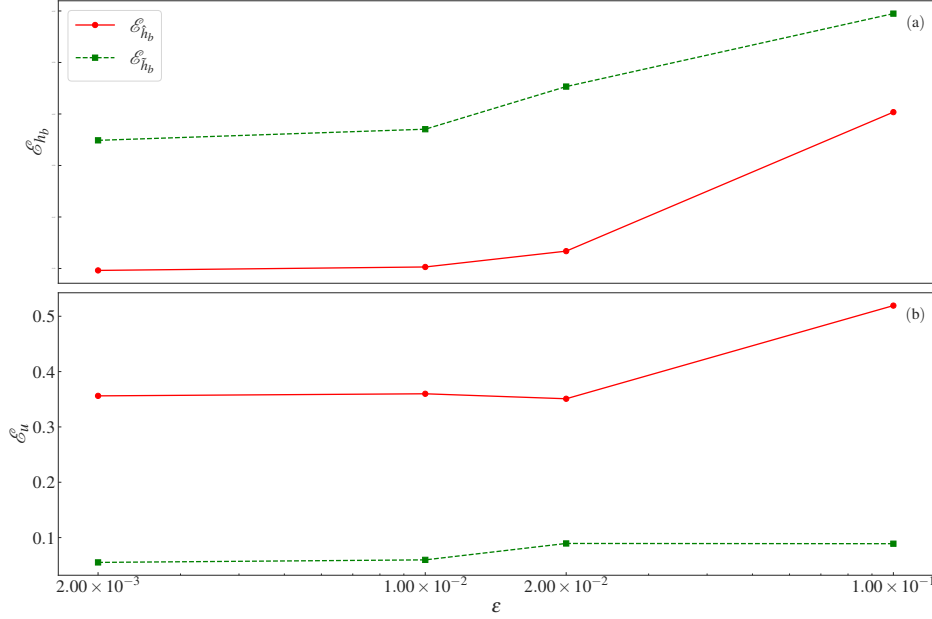


Figure 8: Errors as function of the added noise ϵ/η_0 . (a) Global error $\mathcal{E}_{h_b}$ (red solid line) and $\mathcal{E}_{h_b}$ (green dotted line), and (b) Global error $\mathcal{E}_u$ (red solid line) and $\mathcal{E}_u$ (green dotted line).

- Flow over an espresso cup: inferring 3-d velocity and pressure fields from tomographic background oriented schlieren via physics-informed neural networks. , 915. Retrieved from <https://www.cambridge.org/core/journals/journal-of-fluid-mechanics/article/abs/flow-over-an-espresso-cup-inferring-3d-velocity-and-pressure-fields-from-tomographic-background-oriented-schlieren-via-physicsinformed-neural-networks/160E4A836637FE3996610389666DA030> (Publisher: Cambridge University Press) doi: 10.1017/jfm.2021.135
- Chiocci, F., Cattaneo, A., & Urgeles, R. (2011). Seafloor mapping for geohazard assessment: state of the art. *Mar Geophys Res*, 32, 1–11. doi: <https://doi.org/10.1007/s11001-011-9139-8>
- Clark Di Leoni, P., Agarwal, K., Zaki, T., Meneveau, C., & Katz, J. (2023). Reconstructing turbulent velocity and pressure fields from under-resolved noisy particle tracks using physics-informed neural networks. *Exp Fluids*, 64(95). doi: <https://doi.org/10.1007/s00348-023-03629-4>
- Clark Di Leoni, P., Agasthya, L., Buzzicotti, M., & Biferale, L. (2023). Reconstructing rayleigh-bénard flows out of temperature-only measurements using physics-informed neural networks. , 46(3), 16. Retrieved from <https://doi.org/10.1140/epje/s10189-023-00276-9> doi: 10.1140/epje/s10189-023-00276-9
- Cuomo, S., Di Cola, V. S., Giampaolo, F., Rozza, G., Raissi, M., & Piccialli, F. (2022). Scientific machine learning through physics-informed neural networks: Where we are and what's next. , 92(3), 88. Retrieved 2022-09-14, from <https://doi.org/10.1007/s10915-022-01939-z> doi: 10.1007/s10915-022-01939-z
- Degueldre, H., Metzger, J. J., Geisel, T., & Fleischmann, R. (2016). Random focusing of tsunami waves. *Nature Physics*, 12, 259–262. doi: <https://doi.org/10.1038/nphys3557>
- Dimet, F.-x. L., & Talagrand, O. (1986). Variational algorithms for analysis and

- assimilation of meteorological observations: theoretical aspects. *Tellus A*, 38A(2), 97-110. Retrieved from <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1600-0870.1986.tb00459.x> doi: <https://doi.org/10.1111/j.1600-0870.1986.tb00459.x>
- Du, Y., Wang, M., & Zaki, T. (2023). State estimation in minimal turbulent channel flow: A comparative study of 4dvar and pinn. *International Journal of Heat and Fluid Flow*, 99, 109073. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0142727X22001412> doi: <https://doi.org/10.1016/j.ijheatfluidflow.2022.109073>
- Eakins, B., & Sharman, G. (2007, 12). Volumes of the world's oceans from etopo1. *AGU Fall Meeting Abstracts*, 0999-.
- Gaggioli, E., & Bruno, O. P. (2022). Parallel inverse-problem solver for time-domain optical tomography with perfect parallel scaling. *Journal of Quantitative Spectroscopy and Radiative Transfer*, 290, 108300. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0022407322002357> doi: <https://doi.org/10.1016/j.jqsrt.2022.108300>
- Glenn, M. F. (2015, Jun.). Introducing an operational multi-beam array sonar. *The International Hydrographic Review*, 47(1). Retrieved from <https://journals.lib.unb.ca/index.php/ihr/article/view/23921>
- Gula, J., Moalemaker, M. J., & McWilliams, J. C. (2015). Gulf stream dynamics along the southeastern u.s. seaboard. *Journal of Physical Oceanography*, 45(3), 690 - 715. Retrieved from <https://journals.ametsoc.org/view/journals/phoc/45/3/jpo-d-14-0154.1.xml> doi: 10.1175/JPO-D-14-0154.1
- Hasanuzzaman, G., Eivazi, H., Merbold, S., Egbers, C., & Vinuesa, R. (2022). Enhancement of PIV measurements via physics-informed neural networks. Retrieved from <http://iopscience.iop.org/article/10.1088/1361-6501/aca9eb> doi: 10.1088/1361-6501/aca9eb
- Irish, J., & White, T. (1998). Coastal engineering applications of high-resolution lidar bathymetry. *Coastal Engineering*, 35(1), 47-71. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0378383998000222> doi: [https://doi.org/10.1016/S0378-3839\(98\)00022-2](https://doi.org/10.1016/S0378-3839(98)00022-2)
- Li, Y., Zhang, J., Dong, G., & Abdullah, N. S. (2020). Small-scale reconstruction in three-dimensional kolmogorov flows using four-dimensional variational data assimilation. , 885, A9. doi: 10.1017/jfm.2019.960
- Mayer, L. (2006). Frontiers in seafloor mapping and visualization. *Mar Geophys Res*, 27, 7-17. doi: <https://doi.org/10.1007/s11001-005-0267-x>
- Mayer, L., Jakobsson, M., Allen, G., Dorschel, B., Falconer, R., Ferrini, V., ... Weatherall, P. (2018, 02). The nippon foundation—gebco seabed 2030 project: The quest to see the world's oceans completely mapped by 2030. *Geosciences*, 8, 63. doi: 10.3390/geosciences8020063
- Pedlosky, J. (1987). *Geophysical fluid dynamics*. Springer Verlag.
- Plessix, R.-E. (2006, 11). A review of the adjoint-state method for computing the gradient of a functional with geophysical applications. *Geophysical Journal International*, 167(2), 495-503. Retrieved from <https://doi.org/10.1111/j.1365-246X.2006.02978.x> doi: 10.1111/j.1365-246X.2006.02978.x
- Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2017). Physics informed deep learning (part I): data-driven solutions of nonlinear partial differential equations. *CoRR*, abs/1711.10561. Retrieved from <http://arxiv.org/abs/1711.10561>
- Sandwell, D. T., Smith, W. H., Gille, S., Kappel, E., Jayne, S., Soofi, K., ... Géli, L. (2006). Bathymetry from space: Rationale and requirements for a new, high-resolution altimetric mission. *Comptes Rendus. Géoscience*, 338(14-15), 1049-1062. doi: 10.1016/j.crte.2006.05.014
- Santoso, A., Cai, W., England, M. H., & Phipps, S. J. (2011). The role of the indonesian throughflow on enso dynamics in a coupled climate model. *Journal of Climate*, 24(3), 585 - 601. Retrieved from <https://journals.ametsoc.org/>

- view/journals/clim/24/3/2010jcli3745.1.xml doi: 10.1175/2010JCLI3745.1
- Shachak Pe’eri, C. A. L. A., Christopher Parrish, & Armstrong, A. (2014). Satellite remote sensing as a reconnaissance tool for assessing nautical chart adequacy and completeness. *Marine Geodesy*, 37(3), 293–314. Retrieved from <https://doi.org/10.1080/01490419.2014.902880> doi: 10.1080/01490419.2014.902880
- Sitzmann, V., Martel, J. N. P., Bergman, A. W., Lindell, D. B., & Wetzstein, G. (2020). *Implicit neural representations with periodic activation functions*. Retrieved from <https://arxiv.org/abs/2006.09661>
- Smith, W., Marks, K., & Schmitt, T. (2017, 03). Airline flight paths over the unmapped ocean. *Eos Transactions American Geophysical Union*. doi: 10.1029/2017EO069127
- Theberge, C. A. E. (1989). sounding to pole beam. *Surveying and Cartography*, 5, 334–346.
- Wang, M., Wang, Q., & Zaki, T. A. (2019a). Discrete adjoint of fractional-step incompressible navier-stokes solver in curvilinear coordinates and application to data assimilation. , 396, 427–450. Retrieved from <http://www.sciencedirect.com/science/article/pii/S0021999119304735> doi: 10.1016/j.jcp.2019.06.065
- Wang, M., Wang, W., & Zaki, T. A. (2019b). Discrete adjoint of fractional-step incompressible navier-stokes solver in curvilinear coordinates and application to data assimilation. *Journal of Computational Physics*, 396, 427–450. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0021999119304735> doi: <https://doi.org/10.1016/j.jcp.2019.06.065>
- Wang, M., & Zaki, T. A. (2021). State estimation in turbulent channel flow from limited observations. , 917. (Publisher: Cambridge University Press) doi: 10.1017/jfm.2021.268
- Wang, Q., Hasegawa, Y., & Zaki, T. A. (2019). Spatial reconstruction of steady scalar sources from remote measurements in turbulent flow. , 870, 316–352. Retrieved from <https://www.cambridge.org/core/journals/journal-of-fluid-mechanics/article/spatial-reconstruction-of-steady-scalar-sources-from-remote-measurements-in-turbulent-flow/C83D9706C0B20578F1331BF614413BBE> (Publisher: Cambridge University Press) doi: 10.1017/jfm.2019.241
- Wang, S., Teng, Y., & Perdikaris, P. (2021). Understanding and mitigating gradient flow pathologies in physics-informed neural networks. *SIAM Journal on Scientific Computing*, 43(5), A3055–A3081. doi: 10.1137/20M1318043
- Wöfl, A.-C., Snaith, H., Amirebrahimi, S., Devey, C. W., Dorschel, B., Ferrini, V., ... Wigley, R. (2019). Seafloor mapping – the challenge of a truly global ocean bathymetry. *Frontiers in Marine Science*, 6. Retrieved from <https://www.frontiersin.org/journals/marine-science/articles/10.3389/fmars.2019.00283> doi: 10.3389/fmars.2019.00283